BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: Physics Paper–X (02) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Physics Paper–X (02) (23)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Time period is reciprocal of.....

(A) Wavelength
(B) Frequency
(C) Displacement
(D) Amplitude

Q2. At 20C°, the speed of sound in air is.....

(A) 343 ms^{-1}
(B) 338 ms^{-1}
(C) 331 ms^{-1}
(D) 330 ms^{-1}

Q3. Sound cannot travel through.....

(A) Solid
(B) Liquid
(C) Gas
(D) Vacuum

Q4. Which of the given materials will refract light more?

(A) Glass
(B) Water
(C) Diamond
(D) Benzene

Q5. One kilowatt-hour is equal to.....

(A) 36 MJ
(B) 3.6 MJ
(C) 0.36 MJ
(D) 360 MJ

Q6. The index of refraction for vacuum is.....

(A) 2.4 (B) 1.5

(C) 1.33 (D) 1

Q7. Charge is measured in the units of.....

(A) Coulomb (B) Ampere

(C) Volt (D) Farad

Q8. The product of current and voltage is....

(A) Resistance (B) Voltage

(C) Power (D) Energy

Q9. A device that converts electrical energy into mechanical energy is.....

(A) D.C motor (B) A.C generator

(C) Transformer (D) Electron gun

Q10. Which of the given is a negatively charged particle?

(A) Proton (B) Alpha particle

(C) Neutron (D) Electron

Q11. Electric or electromagnetic representation of data or information is termed as.....

(A) Software (B) Signal

(C) Transmission (D) Computer

Q12. Origin of energy from the sun and stars is.....

(A) Radioactivity (B) Fission

(C) Fusion (D) Beta emission

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Answer any EIGHT parts. Each part carries FOUR marks.

Q1. Differentiate between transverse and longitudinal waves.

Q2. What do you mean by audible frequency range?

Q3. Why sound cannot be heard on moon?

Q4. Define power of lens and its unit.

Q5. Define electrical resistance. What are the factors upon which it depends?

Q6. If the speed of light in Kerosene oil is $2.0^8 \times 108$ m/s, calculate the index of refraction.

Q7. Define the unit of electrical energy. Show that 1 KWh = 3.6 MJ.

Q8. Differentiate between electric and magnetic fields. 0.055

Q9. What is the function of an accelerating anode in an electron gun?

Q10. Briefly explain e-mail and internet.

Q11. Write down four applications of radioisotopes.

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

Q12.

(a) [4] Explain the terms echo and reverberation. Calculate the minimum distance for echo to be heard.

(b) [3] The P.D across the plates of a capacitor is 500 V, the charge on each plate is 0.02 C. What is the capacitance of the capacitor?

Q13.

(a) [4] What is series combination of capacitors? Determine the equivalent capacitance for three capacitors C_1, C_2 and C_3 connected in series.

(b) [3] Calculate the period and frequency of a propeller on a plane if it completes 250 cycles in 5 s.

Q14.

(a) [4] Describe the construction and working of AC generator.

(b) [3] What is the resistance of wire when the current through it is 2.0 A and the voltage across its ends is 3.0 V?

Q15.

(a) [4] Define and explain half life of a radioactive element. 0.055

(b) [3] How much work is done in moving a charge of 3 C from a point at 118 V to a point at 138 V in an electric field?